Aarush Dhuri

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EDUCATION

Rahul International School — **Mumbai, India**
12th Grade – 74.02% — 2022 – 2024

NMIMS Deemed-to-be-University, MPSTME — **Mumbai, India**
B.Tech in Data Science – CGPA: 8.72/10 — 2024 – 2028

PROJECTS

Brain Battery Check – Mental Health Score Prediction Platform | *Python, Scikit-learn, FastAPI, Pandas* **GitHub** 2026

- Developed a machine learning platform that predicts student mental health scores using social media usage, academic habits, sleep patterns, physical activity, and stress indicators.
- Built an end-to-end ML pipeline including data preprocessing, country grouping, feature engineering, logarithmic transformations, scaling, ordinal encoding, and one-hot encoding for robust model training.
- Trained and evaluated Linear Regression and Random Forest models, improving performance through hyperparameter tuning with RandomizedSearchCV and automated model selection.
- Designed and deployed a FastAPI-based prediction service with Pydantic validation, enabling real-time mental health score estimation through REST APIs.
- Implemented a reusable production-ready inference pipeline using Joblib serialization and integrated support for multiple demographic, academic, and behavioral features.

Multi-Agent PR Review System | *Python, FastAPI, React, Ollama, SQLite* **GitHub** | Sep 2026 – Oct 2026

- Developed an AI-powered code review system with five specialized agents for security vulnerability detection, bug identification, performance analysis, testing recommendations, and documentation review.
- Integrated the GitHub API to fetch pull request metadata and code diffs for automated analysis.
- Integrated a locally hosted Qwen3:8B model through Ollama and built a deterministic aggregation mechanism to consolidate findings into a unified review report.
- Deployed backend REST endpoints delivering fast, reliable inferences with consistent sub-500ms response times for review workflows.
- Developed a React frontend and FastAPI backend with SQLite-based review history, enabling users to submit pull request URLs and view analysis results.

RAG-Based Video-Speech Assistant | *Python, Sarvam AI, LangChain, ChromaDB, LLMs* Jun 2026 – Mar 2026

- Developed a video-speech assistant integrating speech transcription, transcript summarization, and action-item extraction.
- Integrated Sarvam AI for speech processing and built LLM-powered workflows for summarizing transcripts and extracting actionable insights.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB for vector storage and context-aware retrieval from processed content.

TECHNICAL SKILLS

Languages: Python, C++, SQL, HTML/CSS

ML & DS: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Supervised & Unsupervised Learning, Regression, Classification, Clustering, PCA, Feature Engineering, Model Evaluation, Statistical Analysis

Artificial Intelligence: Generative AI, Large Language Models (LLMs), Prompt Engineering, LangChain, Retrieval-Augmented Generation (RAG)

Cloud & DevOps: AWS (Amazon Bedrock, Amazon SageMaker, Amazon S3, Amazon EC2, IAM, AWS Lambda, CloudWatch), Docker, Docker Compose, Git, GitHub

ACHIEVEMENTS

Hackathons: Participated in Smart India Hackathon 2025; Certificate of Merit.

Certifications: AWS Certified AI Practitioner, RAG Systems Agentic Workflows with Pinecone and LangGraph (Coursera), AI for Everyone (DeepLearning.AI)